

Expressibility–Trainability Tradeoffs in Noisy Parameterized Quantum Circuits: A Crossed Feature-Map and Ansatz Study

Sounak Bhowmik
Southern Methodist University, Dallas
sbhowmik@smu.edu

Abstract—Selecting a quantum feature map, trainable ansatz, circuit depth, and entanglement topology jointly determines both the state-space coverage and the optimization difficulty of a quantum machine-learning model. In this paper, we study this interaction with a crossed design comprising three feature maps, two ansatz families, three circuit widths, four depths, and three connectivity patterns. We calibrated the experiments in noisy and noiseless simulation, and ten random seeds. Expressibility is measured by the Kullback–Leibler divergence between sampled state fidelities and the Haar distribution; entangling capability by the Meyer–Wallach measure; and trainability by the variance of a parameter-shift gradient of a data-encoded, composed feature-map–ansatz circuit. Across the available noiseless configurations, mean gradient variance at ten qubits decreases from 0.123 at one repetition to 0.0496 at eight repetitions. The decrease is concentrated in the entangling `real_amplitudes` ansatz: for linear ten-qubit circuits, its three feature-map combinations lose 92.5–99.3% of gradient variance by depth eight, whereas the nonentangling `rotation_only` ansatz remains substantially more trainable. Hardware-calibrated noise produces an even sharper topology effect. At ten qubits, the median retained gradient variance for linear circuits changes from 0.916 at depth one to 0.545 at depth eight, while most circular and fully connected circuits collapse toward zero after routing to the sparse device patch. These results support joint, hardware-aware selection of encoding, ansatz, depth, and topology, but do not support a universal feature-map-independent saturation depth.

Index Terms—quantum machine learning, parameterized quantum circuits, expressibility, trainability, barren plateaus, noise-aware compilation

I. Introduction

Parameterized quantum circuits are central to near-term variational algorithms, but their design presents a tension. Greater state-space coverage can improve representational capacity, while sufficiently random or expressive circuits can concentrate objective values and suppress gradients [1], [2]. At the same time, near-term processors impose gate errors and sparse connectivity, making nominally compact all-to-all circuits costly after routing [3]. Circuit selection must therefore consider expressibility, trainability, and hardware compatibility together.

Quantum feature maps embed classical samples in Hilbert space and underlie both quantum-kernel and variational-classifier constructions [4]. Prior expressibility studies introduced fidelity-based and entanglement-based

descriptors and showed that both can saturate with circuit depth [5]. The present study extends this circuit-level perspective in three practical directions. It (i) crosses feature maps and ansätze instead of pairing them one-to-one, (ii) measures ansatz trainability after composing it with a feature map bound to a real data sample, and (iii) evaluates the same design under a fixed, hardware-calibrated noise model and qubit patch.

The main contributions are as follows:

- A factorial comparison of three encoding circuits and two structurally distinct ansätze over width, depth, topology, and noise.
- A reproducible measurement protocol linking feature-map expressibility to the gradient variance of the composed classifier circuit.
- Evidence that depth-induced loss of trainability depends strongly on ansatz structure, and that routing topology can dominate this loss under calibrated noise.

II. Metrics and Problem Formulation

A. Expressibility

Following Sim et al. [5], two independent parameter vectors are sampled for a feature map $U(\theta)$, and the fidelity $F = |\langle \psi(\theta) | \psi(\phi) \rangle|^2$ is recorded. For a Hilbert-space dimension $N = 2^n$, the Haar fidelity density is

$$P_{\text{Haar}}(F) = (N - 1)(1 - F)^{N-2}. \quad (1)$$

Expressibility is the histogram estimate

$$E_{\text{KL}} = D_{\text{KL}}(P_{\text{circuit}} \parallel P_{\text{Haar}}). \quad (2)$$

A smaller E_{KL} denotes a more Haar-like, and hence more expressible, circuit ensemble.

B. Entangling Capability

For each random parameter draw, the Meyer–Wallach global-entanglement measure [6] is computed as

$$Q = 2 \left(1 - \frac{1}{n} \sum_{k=1}^n \text{Tr}(\rho_k^2) \right), \quad (3)$$

where ρ_k is the one-qubit reduced state. The reported capability is the sample mean of Q , ranging from zero for product states to one for maximally entangled single-qubit marginals.

C. Trainability

The trainability proxy is the variance, over random ansatz initializations, of one component of the gradient of the local observable Z_0 . The first trainable parameter is differentiated by the parameter-shift rule [7]:

$$g(\theta) = \frac{f(\theta + \frac{\pi}{2}\mathbf{e}_1) - f(\theta - \frac{\pi}{2}\mathbf{e}_1)}{2}, \quad T = \text{Var}_\theta[g]. \quad (4)$$

Larger T indicates gradients that survive random initialization; values near zero indicate an effectively flat landscape. Unlike the other two metrics, T is evaluated on the feature map followed by the ansatz. The feature-map parameters are bound to a training sample, leaving only the ansatz parameters free.

For comparing noise conditions, retained trainability is

$$R_T = \frac{T_{\text{noisy}}}{T_{\text{noiseless}}}. \quad (5)$$

A value near one indicates little noise-induced loss; a value near zero indicates that noise has erased parameter sensitivity.

III. Experimental Methodology

A. Data Preparation

The experiment uses the 569-example, 30-feature Breast Cancer Wisconsin (Diagnostic) binary-classification dataset [8]. A stratified 80/20 split gives 455 training and 114 test examples. The implementation standardizes the raw training features, fits principal-component analysis to the selected qubit count, standardizes the retained components again, and multiplies them by $\pi/2$ to obtain encoding angles. Every transform is fit only on the training split. Although a test set is prepared, the current experiment reports circuit descriptors rather than classifier test accuracy.

B. Crossed Circuit Design

The feature-map families are:

- `local_z`: repeated single-qubit Z rotations without entanglement;
- `zz`: a diagonal, commuting ZZ feature map; and
- `cx_ry`: data-dependent R_y rotations followed by CX entanglers.

The same data parameters are reused at every repetition. Each feature map is crossed with (i) `rotation_only`, containing repeated R_y layers and no entanglers, and (ii) `real_amplitudes`, containing R_y layers and CX entanglers. The crossed design separates the encoding effect from the ansatz effect more clearly than fixed feature-map-ansatz pairing.

The planned grid contains widths $n \in \{4, 8, 10\}$, depths $d \in \{1, 2, 4, 8\}$, and linear, circular, and full entanglement patterns. The topology is immaterial for the entirely local `local_z+rotation_only` circuit, so that pair is evaluated only with the linear label. This gives 192 planned circuit configurations. One result

shard—`local_z+real_amplitudes`, 10 qubits, depth 8, full topology—is absent from the archived results. The analysis therefore uses 191 complete configurations, each observed for ten seeds and two noise conditions, totaling 3,820 raw rows.

C. Simulation and Noise

Noiseless metrics use exact statevector simulation. Per seed, expressibility, entangling capability, and gradient variance use 500, 200, and 200 Monte Carlo samples, respectively. Noisy evaluation uses density matrices and reduced counts of 60, 30, and 30 samples to control runtime. Ten seeds both initialize the Monte Carlo sampling and select the first ten training examples cyclically; consequently, variation across seeds includes both random initialization and data-instance variation.

The implementation uses Qiskit [9] and a noise model derived from the 127-qubit FakeSherbrooke backend. Every circuit is restricted to a nested prefix of physical qubits 0–9, which forms a connected chain in the selected heavy-hex patch. Parameterized circuits are transpiled once with optimization level 3 to the $\{\text{sx}, \text{rz}, \text{x}, \text{ecr}\}$ basis and then simulated with the backend-derived noise model. This fixed patch avoids confounding circuit width with a change of device while exposing routing overhead when circular or full logical connectivity is embedded in sparse hardware.

D. Analysis Protocol

Each reported cell is the mean over ten seeds; plotted error bars are one sample standard deviation. Depth trends are descriptive rather than inferential. Rank correlations use the 12 linear, noiseless cells formed by three feature maps and four depths within each ansatz and width. Missing configurations are not imputed.

IV. Experimental Results

A. Feature-Map Expressibility Is Family Specific

Fig. 1 shows that increasing depth does not produce one universal expressibility curve. At ten qubits with linear connectivity, `cx_ry` improves from $E_{\text{KL}} = 0.536$ at depth one to 2.73×10^{-3} at depth eight. The `zz` family is already Haar-like relative to the other families and remains within 0.045–0.105. By contrast, repeated `local_z` rotations do not expand the reachable product-state family; their estimated divergence increases from 0.364 to 5.79 at depth two before decreasing to 2.81 at depth eight. These results show family-dependent plateaus and nonmonotonic finite-sample behavior, not a common saturation depth.

B. Ansatz Structure Controls Depth-Induced Gradient Loss

Table I isolates the ansatz effect at ten qubits with linear connectivity. Every `real_amplitudes` combination loses more than 92% of its noiseless gradient variance between depths one and eight. The strongest collapse

Where expressibility saturates (10 qubits, linear topology, noiseless)

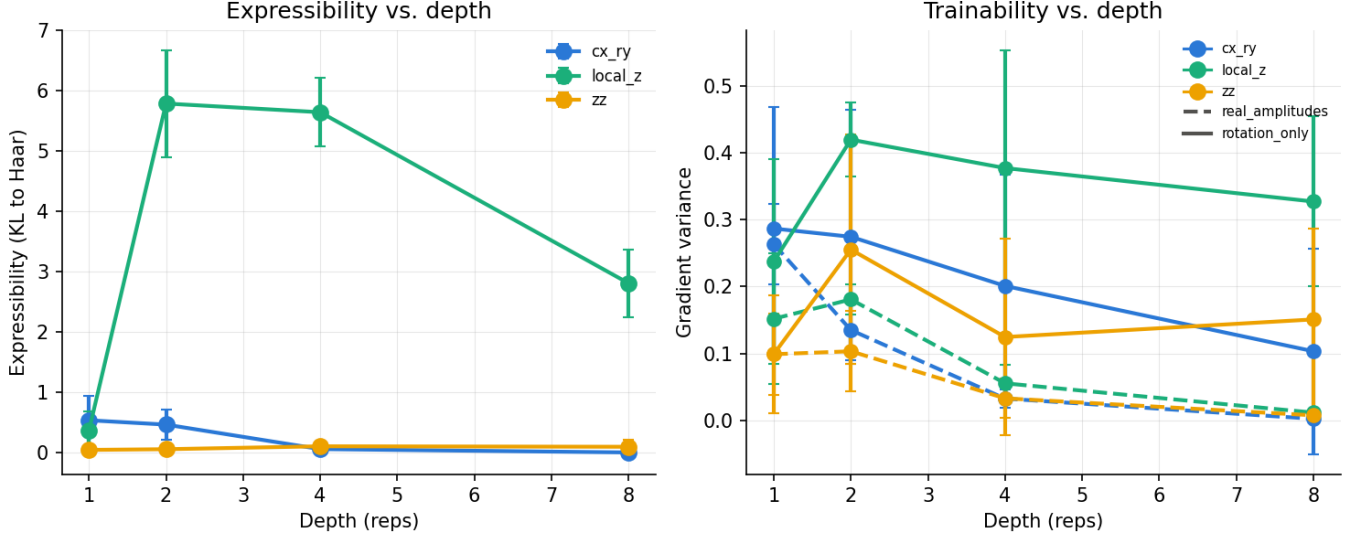


Fig. 1. Depth dependence in the noiseless, ten-qubit linear setting. Lower KL divergence means greater expressibility; larger gradient variance means better trainability. Solid and dashed trainability curves denote rotation_only and real_amplitudes, respectively.

TABLE I
Noiseless Gradient-Variance Endpoints (10 Qubits, Linear)

Feature map	Ansatz	$T_{d=1}$	$T_{d=8}$	Change
cx_ry	real amp.	0.2637	0.0020	-99.3%
local_z	real amp.	0.1518	0.0113	-92.6%
zz	real amp.	0.0987	0.0072	-92.7%
cx_ry	rotation	0.2870	0.1032	-64.0%
local_z	rotation	0.2373	0.3275	+38.0%
zz	rotation	0.0988	0.1509	+52.7%

occurs for cx_ry, from 0.264 to 1.98×10^{-3} (99.3%). The nonentangling rotation_only ansatz is more resistant: its cx_ry combination loses 64.0%, while its local_z and zz combinations finish above their depth-one values, albeit nonmonotonically.

After averaging over all available feature-map, ansatz, and topology cells, ten-qubit noiseless gradient variance decreases from 0.123 at depth one to 0.0496 at depth eight, a 59.8% reduction. The corresponding reduction at four qubits is 36.7% (0.139 to 0.0879), indicating a width-depth interaction. In the linear setting, the descriptive Spearman association between E_{KL} and T is positive, as expected when greater expressibility (smaller KL) accompanies smaller gradients. Across widths 4, 8, and 10, the correlation is 0.273–0.462 for real_amplitudes and 0.860–0.895 for rotation_only. Fig. 2 visualizes these trajectories; their differing shapes caution against reducing trainability to expressibility alone.

C. Sparse Hardware Strongly Penalizes Dense Logical Topology

Fig. 3 reports R_T at ten qubits. For the six linear feature-map-ansatz combinations, the median retention is 0.916, 0.873, 0.824, and 0.545 at depths 1, 2, 4, and 8, respectively. At depth eight, the three rotation_only linear circuits retain 0.593–0.930, whereas the three real_amplitudes circuits retain 0.134–0.498.

The circular and full logical topologies are qualitatively different after routing to the chain-like patch. At ten qubits and depth eight, all three circular real_amplitudes combinations have $R_T < 6 \times 10^{-11}$; the available full-topology combinations are also effectively zero. Even for rotation_only, entanglers in the feature map can induce this routing penalty: depth-eight circular cx_ry retains 7.91×10^{-4} and circular zz is numerically indistinguishable from zero. The result is therefore not simply “noise increases with depth.” It is an interaction between the composed circuit’s entanglers and the mismatch between logical and physical connectivity.

D. Implications for Circuit Selection

The combined results favor a staged configuration policy. First reject topologies that require extensive routing on the target coupling graph. Then compare feature-map expressibility within the remaining shallow circuits, and finally evaluate trainability on the actual composed feature-map-ansatz model. Feature-map descriptors alone are insufficient: the same encoding can sit above a robust rotation-only ansatz or an entangling ansatz whose gradients collapse with depth. Conversely, a nominally

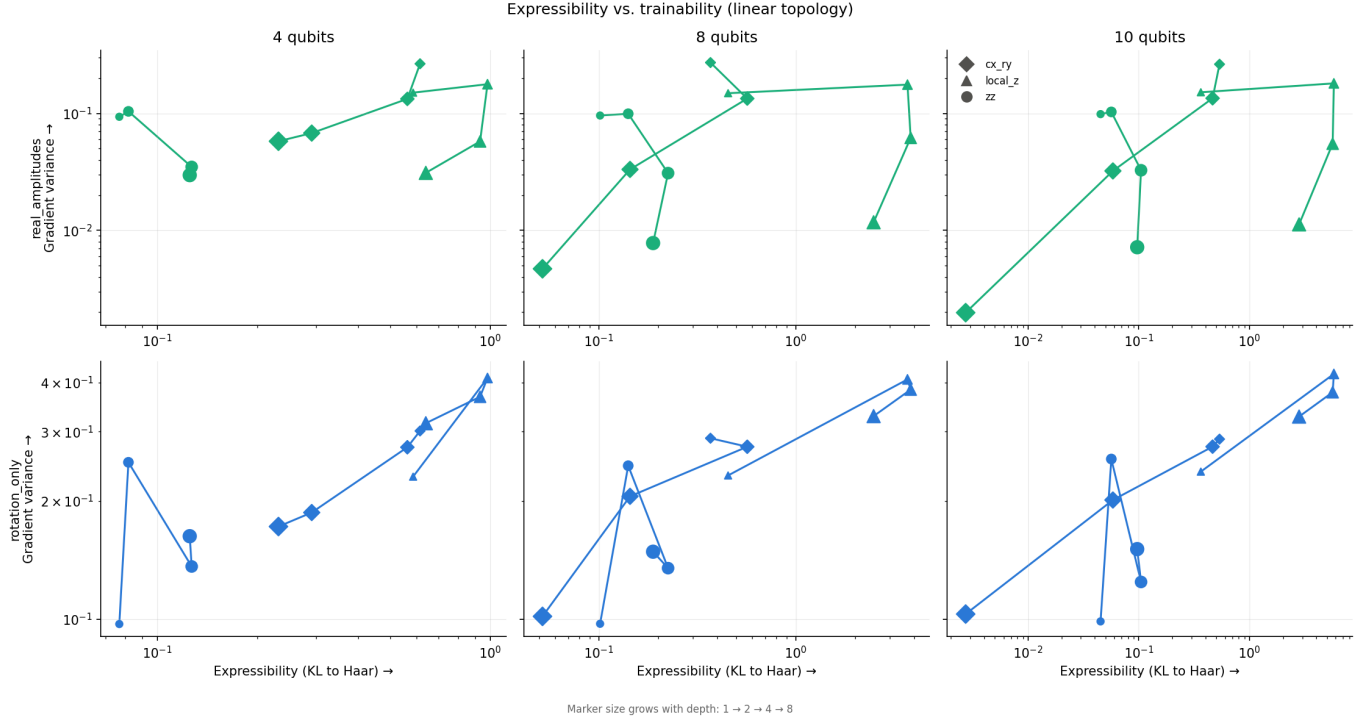


Fig. 2. Noiseless expressibility–trainability trajectories for linear topology. Rows are ansätze, columns are widths, marker shape identifies the feature map, and marker size increases with depth. Both axes are logarithmic.

nonentangling ansatz does not eliminate noise sensitivity if its feature map introduces dense two-qubit interactions.

V. Limitations and Threats to Validity

The experiment is a simulation study on one dataset and one calibrated fake backend; it does not execute circuits on live hardware. Backend calibrations represent a snapshot and results may change across devices or dates. Only one local observable and the first ansatz parameter are used for the gradient statistic, so the metric does not characterize the full optimization landscape. Ten seeds bind ten different training examples, confounding initialization variability with input variability; future work should cross data samples and initialization seeds independently.

No end-to-end classifier is trained, so “trainability” here means gradient survival rather than achieved accuracy, convergence time, or generalization. The noisy Monte Carlo budget is smaller than the noiseless budget, which increases uncertainty in ratios near zero. Exact density-matrix simulation also limits the study to ten qubits. Finally, one of 192 scheduled configurations is missing and is excluded without imputation. Conclusions about full-topology, ten-qubit, depth-eight behavior should therefore be read with that missing `local_z+real_amplitudes` cell in mind.

VI. Conclusion

This crossed experiment shows that expressibility and trainability cannot be assigned to a feature map or

ansatz in isolation. Under noiseless simulation, increasing depth strongly suppresses gradients for the entangling `real_amplitudes` ansatz, while a rotation-only ansatz can remain trainable even at depth eight. Under a fixed calibrated noise model, logical topology becomes decisive: linear circuits retain useful parameter sensitivity, whereas routing circular and fully connected circuits onto a sparse patch often produces near-complete gradient collapse. The practical conclusion is a hardware-aware one: connectivity-compatible shallow circuits should be screened before paying for full variational training, and the screen must evaluate the composed encoding–ansatz circuit rather than a single component.

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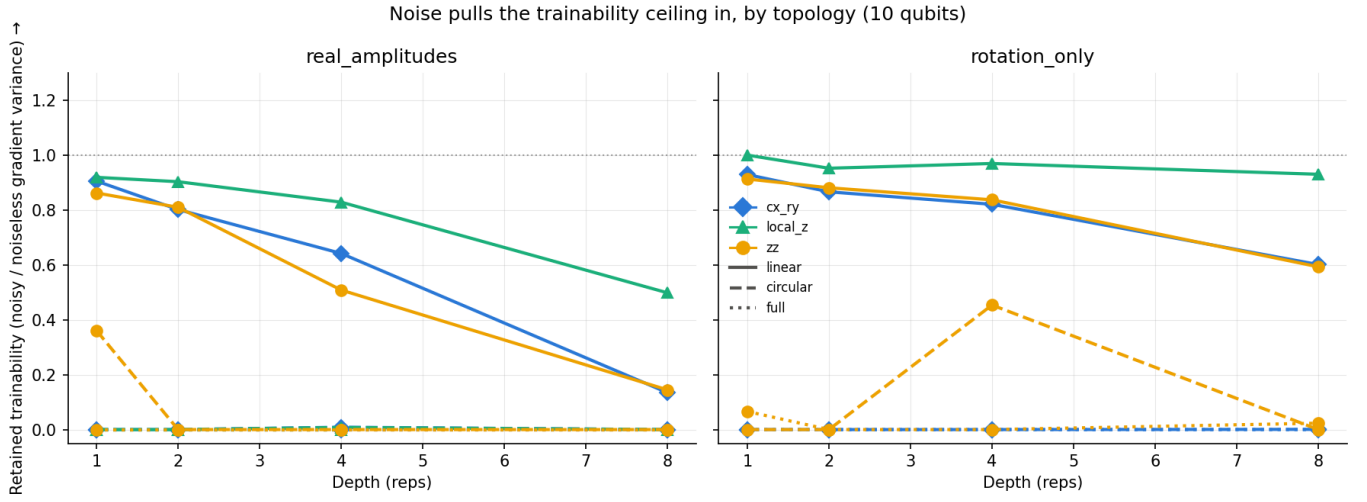


Fig. 3. Retained trainability R_T under the calibrated Sherbrooke noise model. Color and marker identify the feature map; line style identifies logical topology. The dotted horizontal reference is $R_T = 1$.

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